A Final Year Project Report

On

Potato Leaf Detection and Disease Prediction

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ABSTRACT

Potato is one of the most widely grown and economically valuable crops of the world its production is highly susceptible to various leaf diseases such as Early Blight, Late Blight, and Septoria Leaf Spot. The infections are highly contagious and cause serious yield and quality losses when not identified and treated on time. Traditional methods of disease identification depend mainly on manual inspections by farmers or experts, which, besides being very time-consuming, is subjective and mostly gives inaccurate results under large-scale field conditions. In the context of the mentioned shortcomings, this project emphasizes the development of an automated potato leaf disease detection system using techniques of image processing and machine learning.

The proposed system works on digital images of potato leaves as input, followed by some pre-processing at different stages involving noise removal, image segmentation, and colour clustering to segregate the affected area. Colour, texture, and shape are the features that can capture disease-specific patterns using different feature extraction methodologies. Finally, these features will be classified by using machine learning or deep learning models that may include Support Vector Machines, Random Forest, or Convolutional Neural Networks. Among them, CNN-based architectures are very effective since they have inherent capabilities to learn complex visual features directly without human intervention.

The system is trained and tested based on public datasets and self-collected field images to ensure the robustness of the algorithm under all light and environmental conditions. Reliability of the model has been assessed based on the performance evaluation metrics, namely accuracy, precision, recall, and F1-score. Experimental results showed that the proposed approach enabled correct identification and differentiation between multiple potato-leaf diseases with high confidence; hence, it has great potential as a decision support tool for farmers.

The automated detection system developed here will go a long way in early diagnosis and reducing crop loss, minimizing the use of chemical pesticides, hence aiding in attaining sustainable agriculture. It contributes to smart farming by integrating Artificial Intelligence into crop health monitoring and is scalable and cost-effective for enhancing potato crop management.

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CHAPTER 1
INTRODUCTION

Introduction

Potato is one of the most important food crops worldwide, serving as a staple food for millions of people. However, diseases such as Early Blight and Late Blight pose major challenges to potato production, often causing significant yield losses [2]. Early and accurate disease detection is therefore essential for maintaining crop health and improving productivity. Conventional disease detection methods, including manual inspection and laboratory testing, are often time-consuming, costly, and not easily accessible to farmers. To overcome these limitations, this project proposes an intelligent potato leaf disease detection system based on image processing and deep learning techniques [3]. The system analyses visible symptoms on potato leaves, such as discoloration, spots, and texture changes, to automatically identify diseases. A Convolutional Neural Network (CNN) is used to learn disease patterns directly from leaf images [1], enabling accurate classification while eliminating the need for manual feature extraction.

1.1. Purpose of the Study

The primary objective of this research is to develop an efficient and reliable potato leaf disease detection system. The specific objectives are:

- To develop a robust potato leaf detection and image preprocessing pipeline.
- To design and train a CNN model for disease classification.
- To evaluate the model's performance under varying environmental conditions.
- To compare the proposed approach with traditional disease detection methods.
- To support automated and large-scale crop monitoring in smart farming applications.

Significance of the Study

This research contributes to modern agriculture by providing a fast, accurate, and cost-effective solution for potato disease detection. Early identification of diseases enables farmers to take timely corrective actions, reduce crop losses, and improve overall yield. The system also promotes sustainable farming by supporting efficient disease management and minimizing unnecessary pesticide use. Furthermore, it demonstrates the potential of artificial intelligence and computer vision in advancing precision agriculture.

1.2. Project Overview

The proposed system first verifies whether an uploaded image contains a potato leaf using a hybrid approach that combines one TensorFlow model and two YOLO models. A voting mechanism is employed to enhance detection accuracy and reliability. Once a potato leaf is confirmed, a CNN-based model classifies it as Healthy, Early Blight, or Late Blight. The classification is based on visual characteristics such as leaf colour, lesion patterns, and texture variations.

Designed to be practical, scalable, and user-friendly, the system can be integrated into smartphones, web applications, and other digital platforms. By enabling quick and accurate disease diagnosis, it reduces reliance on manual inspection, supports early intervention, and promotes precision agriculture.

CHAPTER 2

LITERATURE OVERVIEW

Literature Overview

Potato is one of the most widely cultivated food crops in the world and serves as a major source of nutrition for millions of people. However, potato plants are highly vulnerable to diseases such as Early Blight and Late Blight, which can spread quickly and cause significant losses in crop yield and quality [2]. Detecting these diseases at an early stage is essential for preventing large-scale damage and ensuring healthy crop production. Over the years, researchers have explored various methods to improve the speed and accuracy of potato disease detection [1].

Traditionally, farmers and agricultural experts identified diseases through visual inspection of plant leaves [2]. Although this method can be effective, it requires experience and can be time-consuming, especially when monitoring large farms. In addition, disease symptoms are sometimes difficult to distinguish with the naked eye, which may lead to incorrect diagnosis. To address these challenges, researchers began using image processing and machine learning techniques to automate the disease detection process [5].

Image Processing Techniques

Image processing plays an important role in preparing leaf images for analysis. Before an image is provided to a machine learning or deep learning model, several preprocessing steps are usually performed to improve image quality and ensure consistent results [3].

Some commonly used techniques include:

- Image resizing to maintain a standard input size.
- Normalization and rescaling of pixel values.
- Data augmentation using rotations, flips, and zooming.
- Noise reduction and image enhancement.
- Segmentation to separate the leaf from the background [3].

These techniques help improve model performance by increasing dataset diversity, reducing overfitting, and making the system more reliable under different environmental conditions.

Machine Learning and Deep Learning Approaches

Early research in plant disease detection mainly relied on traditional machine learning algorithms. In these approaches, features such as colour, texture, shape, and disease spots were manually extracted from leaf images and then used for classification [5].

Some commonly used machine learning methods include:

- Support Vector Machines (SVM)
- Random Forest
- Decision Trees
- k-Nearest Neighbours (k-NN)
- Artificial Neural Networks (ANN) [5]

Although these methods produced satisfactory results in controlled environments, their performance often decreased when images were captured under varying lighting conditions or complex backgrounds.

With the rapid growth of artificial intelligence, deep learning has emerged as a more powerful solution [2]. Convolutional Neural Networks (CNNs) have become

particularly popular because they can automatically learn important features directly from images without requiring manual feature extraction [1]. CNNs are capable of identifying disease-related characteristics such as discoloration, lesions, texture changes, and abnormal patterns with high accuracy [4].

Researchers have successfully applied several CNN architectures, including:

- VGG16 and VGG19
- ResNet
- MobileNet
- DenseNet
- EfficientNet [6]

Many studies have also adopted transfer learning techniques, allowing models to benefit from knowledge gained from large pre-trained datasets. These approaches have significantly improved classification accuracy while reducing training time and computational requirements [7].

Datasets and Evaluation Methods

The effectiveness of a disease detection model largely depends on the quality of the dataset used for training and testing. Researchers commonly use datasets such as PlantVillage, PlantDoc, and field-collected potato leaf datasets [2].

While controlled datasets provide clean and well-labelled images, real-world datasets are more challenging because they contain variations in lighting, background conditions, leaf orientation, and disease severity. Training models on diverse datasets helps improve their ability to perform effectively in practical agricultural environments.

To evaluate model performance, researchers commonly use:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix
- Loss Value

These metrics help determine how accurately and consistently a model can identify different disease categories [4].

Challenges in Potato Leaf Disease Detection

Despite significant advancements in this field, several challenges still remain [2]. Real-world agricultural environments are highly variable, making disease detection more difficult than in controlled laboratory conditions.

Some major challenges include:

- Imbalanced datasets.
- Changes in lighting and weather conditions.
- Complex and cluttered backgrounds.
- Similar symptoms among different diseases.
- Limited availability of labelled data.
- Difficulty in identifying diseases during their early stages.
- Deployment of deep learning models on low-resource devices.

Overcoming these challenges is essential for developing reliable and practical disease detection systems.

Current Trends and Future Directions

Recent research is moving beyond simple disease classification toward intelligent agricultural support systems [2]. Modern studies focus on combining disease detection with additional information sources to provide more useful insights for farmers.

Some important research directions include:

- Integration of weather and environmental data.
- Development of lightweight mobile-friendly models.
- Explainable Artificial Intelligence (XAI).
- Real-time disease monitoring systems.
- Cloud-based agricultural platforms.
- AI-powered treatment recommendation systems.
- Precision farming and smart agriculture solutions [6].

These developments aim to provide farmers with not only disease predictions but also practical recommendations for crop management.

Relevance to the Proposed System

The reviewed literature shows that image processing and deep learning techniques are highly effective for potato leaf disease detection. Building on these findings, the proposed system combines potato leaf detection, CNN-based disease classification, weather integration, and treatment recommendations into a single platform. This approach not only identifies diseases but also provides farmers with useful information that can support timely decision-making and improve crop management. By integrating artificial intelligence with environmental analysis, the system contributes to the development of smarter and more efficient agricultural practices.

CHAPTER 3
PROBLEM DEFINITION & OBJECTIVE

Problem Definition

Potato crops are vulnerable to diseases such as Early Blight and Late Blight, which can cause significant yield losses if not detected in time. Conventional disease identification mainly relies on manual inspection, a process that can be time-consuming, subjective, and prone to errors.

Variations in lighting conditions, leaf appearance, and background complexity further reduce the effectiveness of traditional detection methods. Additionally, these approaches are difficult to scale for large agricultural fields and may not provide timely results. Therefore, there is a need for an automated and reliable system that can accurately detect potato leaf diseases using image processing and deep learning techniques [5].

Objectives

Primary Objective

To develop a fast, accurate, and user-friendly system for detecting potato leaf diseases from images using deep learning techniques, helping farmers take timely action and reduce crop losses.

Technical Objectives

- Develop a CNN-based model to classify potato leaves as Healthy, Early Blight, or Late Blight.
- Achieve high classification accuracy with minimal false predictions.
- Ensure reliable performance under varying lighting conditions, viewing angles, and image qualities.
- Optimize the system for real-time deployment on smartphones and other resource-constrained devices.

Research Objectives

- Evaluate the impact of image preprocessing techniques such as augmentation and normalization.
- Assess model robustness using noisy or partially damaged leaf images.
- Compare the proposed approach with existing disease detection methods.
- Analyse visual symptoms, including spots, discoloration, and texture changes, to improve model interpretability.

Implementation Objectives

- Develop an efficient image preprocessing and prediction pipeline.
- Enable real-time disease detection through web and mobile applications.
- Design a simple interface that displays disease type and relevant recommendations.
- Ensure scalability to support larger datasets, additional diseases, and integration with smart farming platforms.

Performance Metrics

The system will be evaluated using the following metrics:

- Accuracy: Correct classification of disease categories.
- Processing Time: Speed of image analysis and prediction.
- False Positives and False Negatives: Reliability of predictions.
- Scalability: Ability to maintain performance as data volume and usage increase.

CHAPTER 4

FEASIBILITY STUDY

Feasibility Study

The feasibility study evaluates the practicality of developing a **Potato Leaf Detection and Disease Prediction System** using image processing and deep learning. The analysis shows that the proposed system is technically achievable, cost-effective, easy to use, and suitable for real-world agricultural applications. It can help farmers detect diseases early, reduce crop losses, and improve decision-making.

Technical Feasibility

The system can be developed using widely available technologies such as **Python, TensorFlow, PyTorch, and OpenCV**. Public datasets like PlantVillage provide sufficient data for training deep learning models. Cloud platforms and GPUs can be used for model development, while optimization techniques enable deployment on smartphones and other low-resource devices. The system can also be integrated into mobile or web applications without requiring specialized hardware.

Operational Feasibility

The proposed solution is simple and user-friendly, requiring only a smartphone to capture leaf images. Farmers can use the system with minimal training and receive disease predictions in real time. It can operate in both online and offline environments, making it practical even in remote farming areas. The system can also be easily updated and expanded to support additional crops and diseases in the future.

Economic Feasibility

The project is economically viable because most development tools are free and open source. It requires minimal hardware investment and offers a low cost per user after deployment. By enabling early disease detection, the system helps reduce crop losses, lowers dependency on expert consultations and laboratory testing, and improves overall farm productivity, resulting in a high return on investment.

Legal and Ethical Feasibility

The system poses minimal privacy concerns since leaf images generally do not contain personal information. Any collected user data can be managed in compliance with data protection regulations. Ethical use is supported through transparent predictions, clear limitations, and the use of diverse datasets to reduce bias. The system is designed to provide fair and accessible support to farmers regardless of their location or level of expertise.

Overall, the feasibility study confirms that the proposed system is practical, affordable, and suitable for deployment in modern agriculture, with strong potential to improve crop health monitoring and support sustainable farming practices.

Feasibility Summary

Aspect	**Feasibility**	**Key Strengths**	**Potential Challenges**	**Mitigation Strategies**
Technical Feasibility	High	Availability of deep learning frameworks (TensorFlow / PyTorch), accessible datasets, and mobile deployment tools.	High computational cost for model training and need for optimized real-time inference.	Use pre-trained models, cloud-based GPU training, and model optimization (quantization, pruning, TFLite).
Operational Feasibility	High	Easy-to-use interface, mobile compatibility, minimal user expertise required, and real-time predictions.	Training farmers or users to properly capture leaf images for accurate results.	Provide simple tutorials, in-app guidance, and clear image capture instructions.
Economic Feasibility	High	Cost-effective due to open-source tools, minimal hardware requirements, and long-term reduction in crop losses.	Initial development and deployment cost for app/website and training dataset preparation.	Utilize free datasets, opensource tools, and low-cost cloud compute options; deploy lightweight models.
Legal and Ethical Feasibility	High	No sensitive personal data involved; easy compliance basic privacy laws	Ensuring that datasets used follow licensing terms and ethical data	Follow opensource dataset licensing rules and include transparent disclaimers about AI predictions.

Table 4.1

The feasibility study concludes that the proposed system is technically, operationally, economically, and legally viable. With its innovative approach and cost-effectiveness, the project holds strong potential for successful implementation in real-world cybersecurity applications.

CHAPTER 5
SYSTEM ANALYSIS/PROPOSED SCHEME

Detection Model Dataset

Dataset Description

A custom **Potato Leaf Image Dataset** was used to train the detection model to distinguish between potato and non-potato leaves. The dataset consists of **7,024 colour images** divided into two classes:

- **potato_leaf**
- **non_potato_leaf**

The images were collected under varying lighting conditions, angles, colours, and leaf conditions, helping the model learn robust features and perform effectively in real-world environments.

Key Features:

- Contains naturally captured colour images of leaves.
- All images are resized to **256 × 256 pixels** for consistency during training.
- Designed to accurately identify whether a leaf belongs to a potato plant, serving as the first step before disease classification [3].

Dataset Classes

Since the dataset only contains two categories, the task is a simple binary classification problem.

Class Name	Description
potato_leaf	Images showing potato plant leaves.
Non_potato_leaf	Images of leaves from other plant types.

Table 5.1

Having clear class separation helps the model focus on the unique features of potato leaves.

Data Preprocessing

Image Loading and Resizing

The images are loaded using TensorFlow's image_dataset_from_directory() function, which automatically assigns labels based on the folder structure. During preprocessing, all images are resized to **256 × 256 pixels** to maintain consistency across the dataset.

To improve training performance, pixel values are normalized to a **0–1 range** using layers. Rescaling(1.0/255). This normalization helps the model train more efficiently and enhances overall stability.

```
dataset = tf.keras.preprocessing.image_dataset_from_directory(
    '../leaves_data',
    seed=124,
    shuffle=True
    image_size= (constant) BATCH_SIZE: Literal[32]
    batch_size=BATCH_SIZE
    )

Found 7024 files belonging to 2 classes.
```

Figure 5.1

```
resize_and_rescale = tf.keras.Sequential([
  #resize the images to (256X256)
  layers.Resizing(IMAGE_SIZE, IMAGE_SIZE),
  #rescale the pixel values to [0,1] dividing by 255
  layers.Rescaling(1.0/255),
])
```

Figure 5.2

Data Augmentation

To make the model more robust and avoid overfitting, simple augmentation techniques are used:

- Random horizontal and vertical flips
- Slight random rotations

```
#here we are using data augmentation to increase the size of the training dataset
# this helps to prevent overfitting and improve the model's performance

from tensorflow.keras.models import Sequential # type: ignore
from tensorflow.keras import layers # type: ignore

data_augmentation=tf.keras.Sequential([
    layers.RandomFlip("horizontal and_vertical"),
    layers.RandomRotation(0.2)
])
```

Figure 5.3

To improve model generalization, data augmentation techniques are applied, allowing the model to learn from leaves appearing in different orientations and conditions. This helps enhance its performance on real-world images.

Dataset Summary:

- Total Images: **7,024**
- Classes: **2 (potato_leaf and non_potato_leaf)**

Dataset Split:

- **80%** Training Set
- **10%** Validation Set
- **10%** Testing Set

This distribution provides sufficient data for training while reserving enough samples for reliable validation and performance evaluation.

Dataset Analysis

Class Distribution

The dataset is reasonably balanced, with **potato_leaf** and **non_potato_leaf** classes containing a similar number of images. This balanced distribution helps reduce model bias and improves classification performance.

Visual Analysis

A visual inspection of the dataset reveals distinct characteristics between the two classes:

- **Potato leaf** images typically exhibit characteristic venation patterns, textures, and, in some cases, disease-related spots.
- **Non-potato leaf** images display a wide range of shapes, textures, and colours, providing clear visual differences that aid classification.

These variations enable the model to learn discriminative features and accurately distinguish potato leaves from other types of leaves.



Figure 5.4

Visual Analysis

A preliminary visual inspection of the dataset highlights noticeable differences between potato and non-potato leaves in terms of shape, vein structure, and texture patterns. However, some non-potato leaves exhibit similar colours and disease-like features, making classification challenging.

To address these variations, preprocessing and data augmentation techniques are applied, enabling the model to learn subtle distinguishing features and improve classification accuracy.

Prediction Model Dataset

Dataset Description

The prediction model uses a **Potato Leaf Disease Dataset** containing **2,152 RGB images** categorized into three classes:

1. **Potato_healthy**
2. **Potato_early_blight**
3. **Potato_late_blight**

The images were collected from real agricultural environments and include variations in lighting conditions, backgrounds, leaf orientations, and disease severity. These variations make the dataset representative of real-world scenarios and help improve the model's robustness [3].

Key Features:

- Contains colour (RGB) images that preserve important texture and disease-related visual patterns.
- Designed to classify potato leaves into healthy, early blight, or late blight categories.
- Includes a balanced distribution of healthy and diseased samples, supporting effective model training and evaluation [4].

Dataset Classes

The dataset is organized into three meaningful classes, each representing a different leaf condition:

Class ID	Class Name	Description
1	Potato_healthy	Healthy potato leaves with no visible infections or discoloration.
2	Potato_early_blight	Leaves showing small brown circular spots with yellow rings — typical symptoms of early blight
3	Potato_late_blight	Leaves with large, irregular dark patches and damaged tissue, commonly seen in late blight.

Table 5.2

These classes capture both mild and severe disease symptoms, giving the model enough variety to learn from.

Data Preprocessing

Before training, the images are processed to ensure they are uniform and suitable for a neural network. This includes resizing, normalization, and real-time augmentation.

Loading the Images

The images were loaded using TensorFlow's image_dataset_from_directory() function:

```
dataset = tf.keras.preprocessing.image_dataset_from_directory(
    '../leaves_data',
    seed=124,
    shuffle=True,
    image_size=(IMAGE_SIZE,IMAGE_SIZE),
    batch_size=BATCH_SIZE
    )
```

Figure 5.5

Output:
Found 2152 files belonging to 3 classes.

Rescaling and Resizing

All images are resized to a fixed dimension, and pixel values are scaled to the range 0,1:

```
resize_and_rescale = tf.keras.Sequential([
    layers.Resizing(IMAGE_SIZE, IMAGE_SIZE),
    layers.Rescaling(1.0 / 255),
])
```

Figure 5.6

Data Augmentation

To make the model more robust, several augmentation techniques were applied:

```
data_augmentation = tf.keras.Sequential([
    layers.RandomFlip("horizontal_and_vertical"),
    layers.RandomRotation(0.3),
    layers.RandomZoom(0.2),
    layers.RandomContrast(0.2),
    layers.RandomTranslation(0.1, 0.1)
])
```

Figure 5.7

Dataset Analysis
Class Distribution
The dataset is generally well-balanced across the three classes: **Potato_healthy**, **Potato_early_blight**, and **Potato_late_blight**. However, the **Potato_late_blight** class contains slightly more images than the others, which may introduce a minor bias during training.

Handling Class Imbalance

Although the imbalance is minimal, models may tend to favour the majority class. To reduce this effect:

- Data augmentation is applied consistently across all classes.
- The dataset is shuffled before training to eliminate order bias.

These measures help ensure balanced learning and improve the model's ability to accurately classify all three potato leaf conditions.

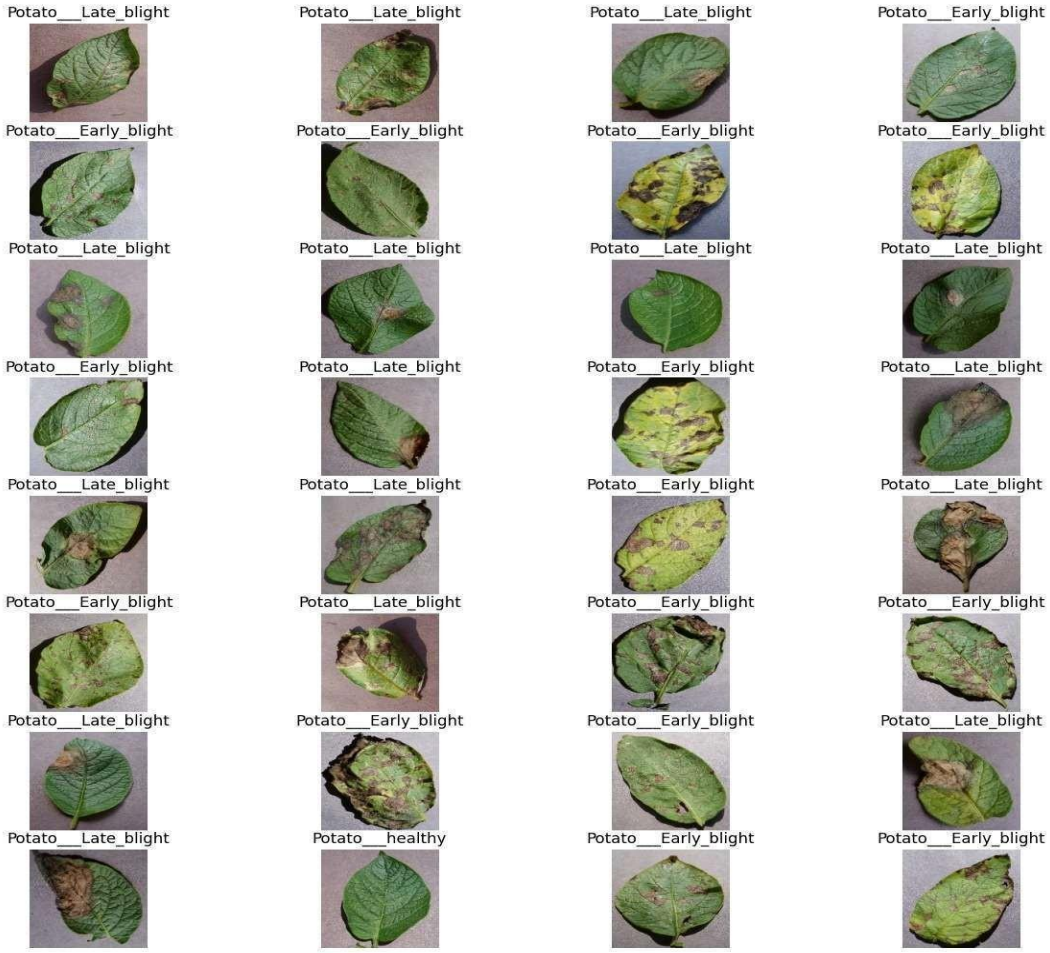


Figure 5.8

Visual Analysis

Visual inspection of the dataset shows distinct characteristics for each class:

- **Healthy leaves** appear uniformly green with smooth and consistent surfaces.
- **Early blight** leaves typically exhibit small dark spots surrounded by yellow halos.
- **Late blight** leaves show larger brown or black lesions, often accompanied by extensive tissue damage.

  Although these differences are generally noticeable, early and late blight symptoms can overlap in some cases, making accurate classification challenging. This emphasizes the need for a CNN-based model capable of learning subtle colour, texture, and pattern variations that may not be easily recognized through manual observation.

CHAPTER 6
Software and Hardware Requirement Specifications

System Design and Software Requirements

1. Introduction

This project provides a mobile-based solution for detecting potato leaves and predicting diseases using images captured from a smartphone. The system consists of two machine learning modules:

1. **Potato Leaf Detection Model** – Determines whether the uploaded image contains a potato leaf.
2. **Potato Leaf Disease Prediction Model** – Identifies the disease affecting the detected potato leaf.

Both models are deployed through separate FastAPI routers, while a React Native (Expo CLI) application serves as the user interface. The objective is to deliver a fast, accurate, and user-friendly tool that supports early disease detection and helps improve agricultural productivity.

2. System Overview

The system operates through the following workflow:

1. The user captures or uploads a leaf image using the mobile application.
2. The image is sent to the FastAPI backend.
3. The backend routes the image to the appropriate detection or prediction service.
4. The selected model processes the image and generates results.
5. The backend returns the output in JSON format.
6. The mobile application displays the results to the user.

This modular architecture allows both services to function independently and scale efficiently.

3. Potato Leaf Detection System

Purpose

The detection module verifies whether the uploaded image contains a potato leaf. This serves as the initial validation step before disease prediction.

Software Requirements

Backend Technologies

- FastAPI
- Uvicorn
- Pydantic

AI/ML Libraries

- TensorFlow
- NumPy
- OpenCV (cv2)
- os

Mobile Frontend

- React Native (Expo CLI)
- Axios / Fetch API

Functional Requirements

- Accept image uploads from the mobile application.
- Detect the presence of a potato leaf.

- Generate a confidence score for detection.
- Return results in JSON format, including leaf status and confidence level.
- Handle invalid, corrupted, or unsupported image files with user-friendly error messages.

4. Potato Leaf Disease Prediction System

Purpose

The prediction module classifies potato leaf images into one of the following categories:

- Healthy
- Early Blight
- Late Blight

A Convolutional Neural Network (CNN) is used to identify disease patterns and perform classification.

Software Requirements

Backend Framework

- FastAPI
- Uvicorn

Machine Learning Libraries

- TensorFlow / Keras
- NumPy
- OpenCV (cv2)

File Management

- os

Mobile Frontend

- React Native (Expo CLI)
- Axios / Fetch API

Functional Requirements

- Accept potato leaf images from the mobile application.
- Perform image preprocessing, including resizing, normalization, and array conversion.
- Predict the disease category and confidence score using the CNN model.
- Return results in JSON format containing the predicted class and confidence value.
- Handle missing files, invalid image data, and unsupported formats effectively.

5. System Integration

Although the detection and prediction modules operate independently, they share a common FastAPI backend, image preprocessing pipeline, JSON-based communication, and mobile interface. This integrated design ensures seamless operation, scalability, and efficient deployment across different platforms.

CHAPTER 7
SYSTEM DESIGN

CNN Model Analysis – Detection Model

Input Layer

The model accepts colour images of size 256 × 256 × 3 (RGB) as input. This allows it to analyse visual characteristics such as colour variations, textures, and leaf patterns.

Feature Extraction Layers

The CNN consists of five convolutional and max-pooling blocks that progressively extract important features from the images.

Layer	Filters	Kernel Size	Activation	Purpose
Conv2D-1	32	3×3	ReLU	Extracts basic features such as edges, colours, and textures
Conv2D-2	64	3×3	ReLU	Learns more complex patterns and leaf structures
Conv2D-3	64	3×3	ReLU	Captures texture and shape information
Conv2D-4	64	3×3	ReLU	Identifies higher-level visual features
Conv2D-5	64	3×3	ReLU	Extracts refined features for classification

Table 7.1

Each convolution layer is followed by a MaxPooling2D (2×2) layer, which reduces the spatial dimensions while preserving important information [4]. After the final pooling layer, the feature map size becomes 6 × 6 × 64.

Flatten Layer

The extracted feature maps are flattened into a one-dimensional vector of 2,304 features, which serves as input to the classification layers [7].

Fully Connected Layers

- Dense Layer: 64 neurons with ReLU activation for learning high-level feature representations.
- Output Layer: 2 neurons with SoftMax activation to generate probability scores for the two classes (potato_leaf and non_potato_leaf).

The CNN architecture combines convolution, pooling, and fully connected layers to effectively learn visual patterns from leaf images [1]. The model is designed to accurately distinguish potato leaves from non-potato leaves while maintaining computational efficiency.

Model Summary

Layer (type)	Output Shape	Param #
sequential (Sequential)	(32, 256, 256, 3)	0
sequential_1 (Sequential)	(32, 256, 256, 3)	0
conv2d (Conv2D)	(32, 254, 254, 32)	896
max_pooling2d (MaxPooling2D)	(32, 127, 127, 32)	0
conv2d_1 (Conv2D)	(32, 125, 125, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(32, 62, 62, 64)	0
conv2d_2 (Conv2D)	(32, 60, 60, 64)	36,928
max_pooling2d_2 (MaxPooling2D)	(32, 30, 30, 64)	0
conv2d_3 (Conv2D)	(32, 28, 28, 64)	36,928
max_pooling2d_3 (MaxPooling2D)	(32, 14, 14, 64)	0
conv2d_4 (Conv2D)	(32, 12, 12, 64)	36,928
max_pooling2d_4 (MaxPooling2D)	(32, 6, 6, 64)	0
flatten (Flatten)	(32, 2304)	0
dense (Dense)	(32, 64)	147,520
dense_1 (Dense)	(32, 2)	130

Figure 7.1

Hyperparameters

The CNN model was trained using the following hyperparameters:

- **Learning Rate:** A learning rate of **0.001** was used in one configuration, while the default learning rate of the Adam optimizer was used in another to enable adaptive learning.
- **Batch Size: 16**, providing a balance between training stability and computational efficiency.
- **Loss Function: Sparse Categorical Cross-Entropy**, suitable for multi-class classification with integer-encoded labels.
- **Optimizer: Adam**, chosen for its adaptive learning capabilities and efficient convergence.
- **Epochs:** The model was trained for **20 epochs** in the first configuration and **30 epochs** in the second, allowing sufficient learning from the dataset.

Training and Testing Splits

Dataset Partitioning

The dataset was divided into:

- **80%** Training Set
- **10%** Validation Set
- **10%** Testing Set

This distribution provides enough samples for model training while ensuring reliable validation and performance evaluation.

Validation Set

The validation set was used during training to monitor model performance, tune parameters, and reduce the risk of overfitting.

Handling Class Imbalance

To address potential class imbalance, techniques such as **data augmentation**, **oversampling**, and **class weighting** were applied when necessary. These methods help ensure that all disease categories are represented fairly during training, improving overall model robustness and prediction accuracy.

```
input_image = (BATCH_SIZE, IMAGE_SIZE, IMAGE_SIZE, CHANNELS)
model = models.Sequential([
    resize_and_rescale,
    data_augmentation,

    layers.Conv2D(32,kernel_size=(3,3), activation='relu', input_shape=input_image),
    layers.MaxPooling2D((2,2)),

    layers.Conv2D(64,kernel_size=(3,3), activation='relu'),
    layers.MaxPooling2D((2,2)),

    layers.Conv2D(64,kernel_size=(3,3), activation='relu'),
    layers.MaxPooling2D((2,2)),


    layers.Conv2D(64,kernel_size=(3,3), activation='relu'),
    layers.MaxPooling2D((2,2)),


    layers.Conv2D(64,kernel_size=(3,3), activation='relu'),
    layers.MaxPooling2D((2,2)),

    layers.Flatten(),
    layers.Dense(64, activation='relu'),
    layers.Dense(len(class_names), activation='sigmoid')
])

model.build(input_shape=input_image)
```

Figure 7.2

CNN Model Analysis – Prediction Model

Input Layer

The prediction model accepts **256 × 256 × 3 RGB images** as input. This enables the network to capture disease-related colour variations, texture patterns, and leaf abnormalities.

Feature Extraction Layers

The CNN architecture consists of **five convolutional and max-pooling blocks** that progressively learn important visual features from potato leaf images [1].

Layer	Filters	Kernel Size	Activation	Purpose
Conv2D-1	32	3 × 3	ReLU	Extracts basic features such as edges, colours, and textures
Conv2D-2	64	3 × 3	ReLU	Learns more detailed disease-related patterns
Conv2D-3	64	3 × 3	ReLU	Captures texture and lesion characteristics
Conv2D-4	64	3 × 3	ReLU	Extracts higher-level disease features
Conv2D-5	64	3 × 3	ReLU	Refines features for classification

Table 7.2

Each convolution layer is followed by a **MaxPooling2D (2 × 2)** layer, which reduces the feature map dimensions while preserving important information [4]. After the final pooling layer, the output feature map size becomes **6 × 6 × 64**.

Flatten Layer

The final feature maps are flattened into a one-dimensional vector containing **2,304 features**, which is then passed to the classification layers [7].

Fully Connected Layers

- Dense Layer**:** 64 neurons with ReLU activation for learning high-level feature representations.
- Output Layer**:** 3 neurons with SoftMax activation, producing probability scores for the three classes:
  - Healthy
  - Early Blight
  - Late Blight

Model_Summary

Layer (type)	Output Shape	Param #
sequential (Sequential)	(32, 256, 256, 3)	0
sequential_1 (Sequential)	(32, 256, 256, 3)	0
conv2d (Conv2D)	(32, 254, 254, 32)	896
max_pooling2d (MaxPooling2D)	(32, 127, 127, 32)	0
conv2d_1 (Conv2D)	(32, 125, 125, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(32, 62, 62, 64)	0
conv2d_2 (Conv2D)	(32, 60, 60, 64)	36,928
max_pooling2d_2 (MaxPooling2D)	(32, 30, 30, 64)	0
conv2d_3 (Conv2D)	(32, 28, 28, 64)	36,928
max_pooling2d_3 (MaxPooling2D)	(32, 14, 14, 64)	0
conv2d_4 (Conv2D)	(32, 12, 12, 64)	36,928
max_pooling2d_4 (MaxPooling2D)	(32, 6, 6, 64)	0
flatten (Flatten)	(32, 2304)	0
dense (Dense)	(32, 64)	147,520
dense_1 (Dense)	(32, 3)	195

Figure 7.3

Hyperparameters

- Learning Rate: 0.001 in one configuration; default Adam learning rate in another.
- Batch Size: 16
- Loss Function: Sparse Categorical Cross-Entropy
- Optimizer: Adam
- Epochs: 20 and 25, depending on the configuration

Training and Testing Splits

- Training Set: 80%
- Validation Set: 10%
- Testing Set: 10%

The **validation set** was used to monitor performance and prevent overfitting. To address potential class imbalance, techniques such as data augmentation, oversampling, and class weighting were applied when required, ensuring balanced learning across all disease categories.

Model Building

```
input_image = (BATCH_SIZE, IMAGE_SIZE, IMAGE_SIZE, CHANNELS)
model = models.Sequential([
    resize_and_rescale,
    data_augmentation,

    layers.Conv2D(32,kernel_size=(3,3), activation='relu', input_shape=input_image),
    layers.MaxPooling2D((2,2)),

    layers.Conv2D(64,kernel_size=(3,3), activation='relu'),
    layers.MaxPooling2D((2,2)),

    layers.Conv2D(64,kernel_size=(3,3), activation='relu'),
    layers.MaxPooling2D((2,2)),


    layers.Conv2D(64,kernel_size=(3,3), activation='relu'),
    layers.MaxPooling2D((2,2)),


    layers.Conv2D(64,kernel_size=(3,3), activation='relu'),
    layers.MaxPooling2D((2,2)),

    layers.Flatten(),
    layers.Dense(64, activation='relu'),
    layers.Dense(len(class_names), activation='softmax')
])

model.build(input_shape=input_image)
```

Figure 7.4

CNN Workflow

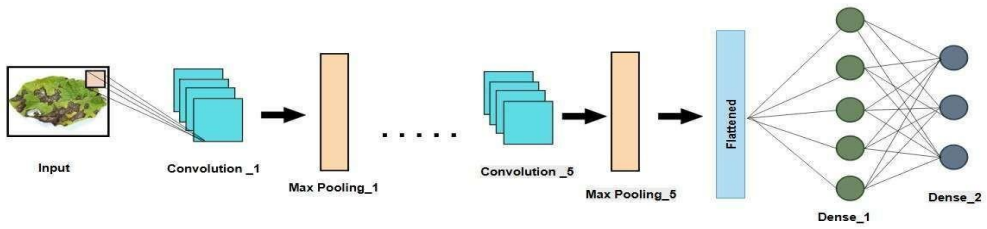


Figure 7.5

CHAPTER 8
RESULTS

Results – Detection Model

The detection model was evaluated using a separate test dataset that was not used during training.

Performance Metrics:

- **Accuracy:** 97.50%
- **Loss:** 0.0552

The high accuracy and low loss indicate that the model can reliably distinguish potato leaves from non-potato leaves with minimal prediction errors. These results demonstrate that the model is accurate, stable, and suitable for real-world deployment in agricultural applications [3].

```
score = model.evaluate(test_ds)
score
```

```
22/22 ━━━━━━━━━━━━━━━━━━━━ 8s 194ms/step - accuracy: 0.9811 - loss: 0.0552

[0.05516644939780235, 0.9811046719551086]
```

Figure 8.1

We are trying to further improve upon our implementation

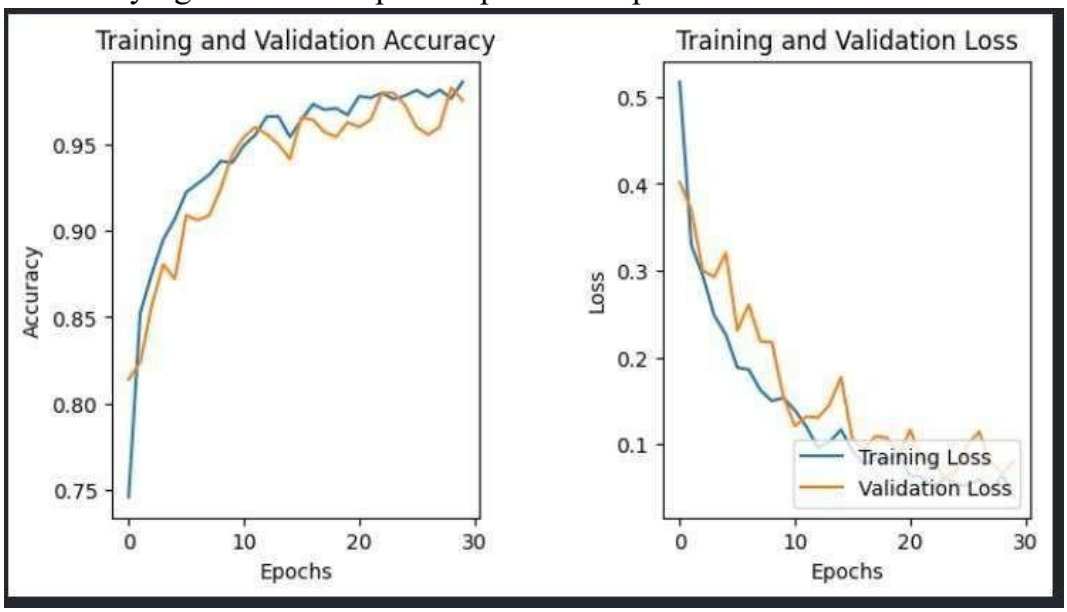


Figure 8.2

Results – Prediction Model

The prediction model was evaluated using a separate test dataset that was not used during training.

Performance Metrics:

- **Accuracy:** 99.14%
- **Loss:** 0.0278

The model correctly classified nearly 99% of the test images, demonstrating its ability to accurately identify potato leaf diseases. The low loss value further indicates strong prediction confidence and minimal classification errors.

These results confirm that the preprocessing techniques and CNN architecture were effective in learning disease-related features, making the system reliable for real-world disease diagnosis and agricultural applications [6].

```
score = model.evaluate(test_ds)
score

8/8 ━━━━━━━━━━━━━━━━━━━━ 2s 165ms/step - accuracy: 0.9914 - loss: 0.0278

[0.027801183983683586, 0.9913793206214905]
```

Figure 8.3

We are trying to further improve upon our implementation.

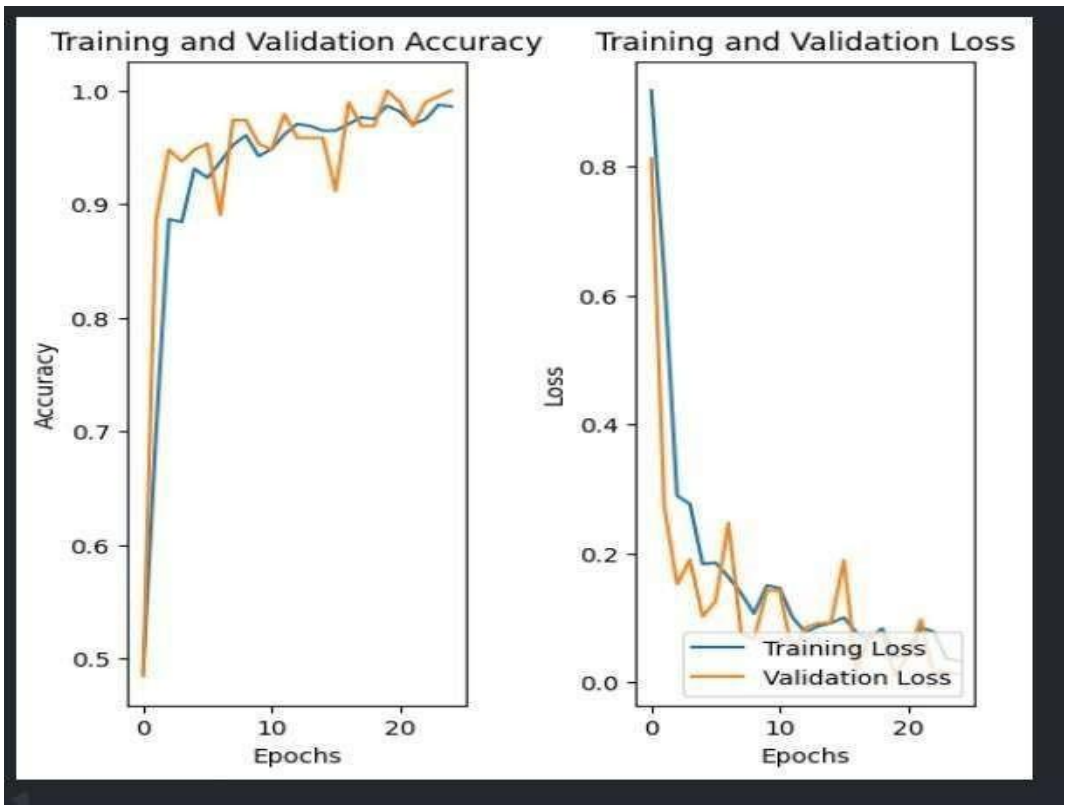


Figure 8.4

Chapter 9
System Workflow

System Workflow

The workflow starts when a user uploads a leaf image through the mobile application. The image is first checked by a YOLO-based model to confirm whether it contains a potato leaf. This step ensures that only valid potato leaf images are processed further.

Once a potato leaf is detected, the image is analysed by a CNN model, which identifies whether the leaf is Healthy, affected by Early Blight, or affected by Late Blight based on its visual characteristics.

After the prediction is made, the application retrieves the user's location and fetches current weather information, such as temperature and humidity. Finally, the system displays the disease prediction along with weather details and treatment recommendations, helping users take timely and informed actions to protect their crops.

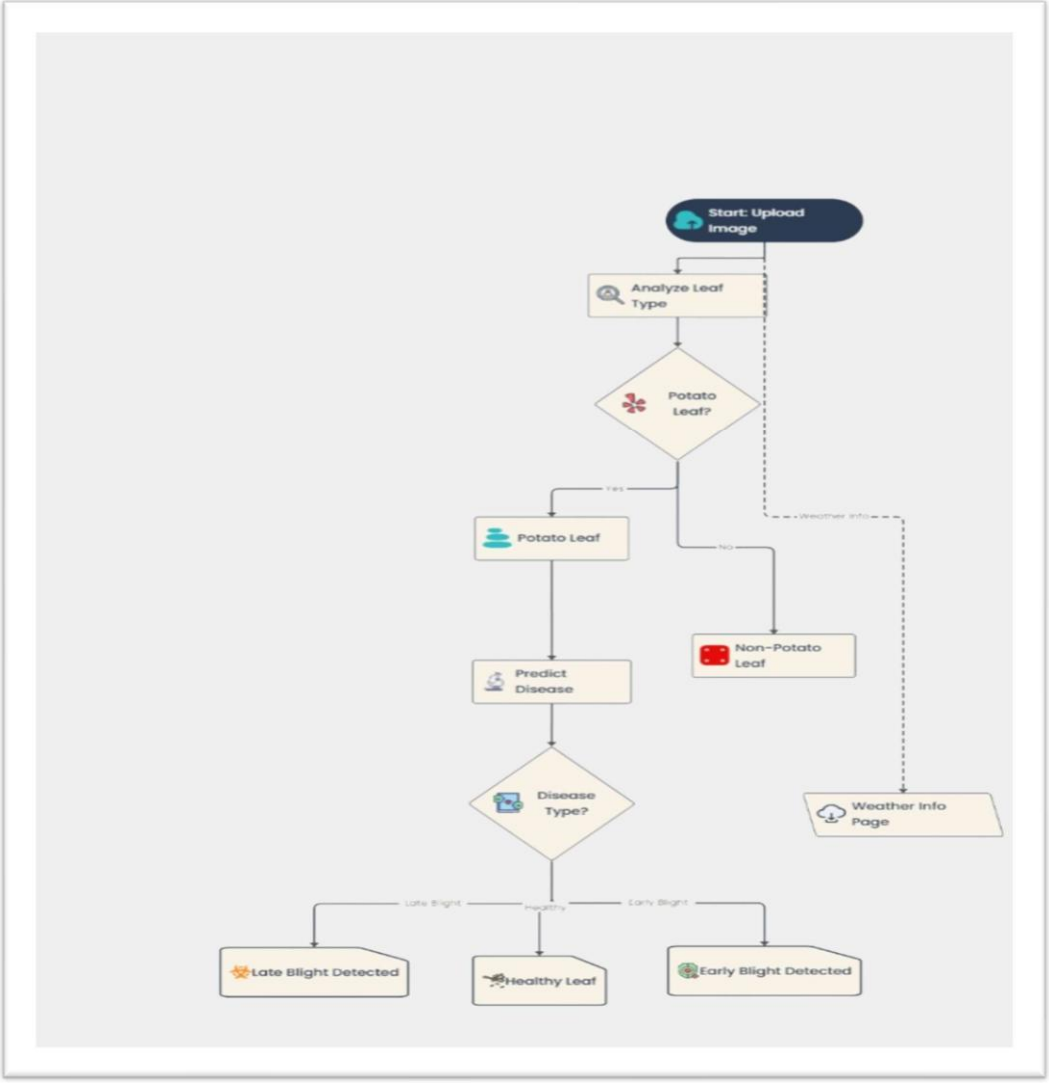


Figure 9.1

Chapter 10
ADDITIONAL FEATURES

Additional Features

1. User Profile

To make the application more personalized and secure, a user profile system is included. Users can create an account, log in securely, and access their previous disease predictions whenever needed. This feature allows farmers to keep track of their crop health and monitor changes over time.

User Authentication

The application uses JWT (JSON Web Token) for authentication and the HS256 algorithm for secure token generation. This ensures that user accounts remain protected and only authorized users can access their information.

Key Features:

- Secure user registration and login.
- Protected user accounts using JWT authentication.
- Safe storage of user credentials.
- Personalized access to application features.

MongoDB Database

All user information, prediction results, and activity records are stored in a MongoDB database. MongoDB provides a flexible and scalable way to manage data efficiently.

Key Features:

- Stores user profiles and account details.
- Maintains disease prediction records.
- Fast and reliable data retrieval.
- Supports future expansion of the application.

User Dashboard

The dashboard acts as the main area where users can view important information related to their account and crop health.

Key Features:

- View profile information.
- Access recent disease predictions.
- Monitor crop health status.
- View weather information and recommendations.
- Easy-to-use and farmer-friendly interface.

Garden and History Tracking

The application keeps a record of previous predictions, allowing users to track the health of their crops over time. This helps identify recurring disease patterns and monitor the effectiveness of treatment measures.

Key Features:

- Prediction history storage.
- Date-wise crop monitoring.
- Tracking of disease occurrences.
- Long-term crop health analysis.

Benefits:

- Helps farmers maintain organized records.
- Makes it easier to monitor crop conditions.
- Supports better decision-making through historical data.

2. Weather Integration

Weather conditions have a significant impact on plant diseases. Certain diseases spread more rapidly under specific temperature and humidity levels [2]. To provide more useful insights, the application integrates real-time weather information using the OpenWeather API.

OpenWeather API Integration

After a disease prediction is made, the application automatically retrieves weather data based on the user's location.

Key Features:

- Real-time weather updates.
- Automatic location-based weather detection.
- Seamless integration with disease prediction results.

Weather Information Display

The system displays important weather parameters that may influence crop health.
Information Provided:

- Temperature
- Humidity
- Weather conditions
- Wind speed
- Atmospheric pressure

Disease Risk Based on Weather

The application analyses weather conditions and provides insights into the potential risk of disease spread. For example, high humidity and moderate temperatures often create favourable conditions for diseases such as Early Blight and Late Blight [6].

Key Features:

- Weather-based disease risk assessment.
- Early warning of favourable disease conditions.
- Better understanding of environmental factors affecting crops.

Benefits:

- Helps farmers take preventive action early.
- Supports better planning of crop management activities.
- Reduces the chances of severe disease outbreaks.

3. Weather-Based Treatment Recommendations

The Weather-Based Treatment Recommendation module enhances the disease prediction system by combining plant disease detection with real-time weather analysis. After a disease is identified, the system retrieves weather information using the OpenWeather API and analyses environmental factors that influence disease development and spread. Based on this analysis, the system calculates a

disease threat score, classifies the risk level, and provides appropriate treatment recommendations to help farmers take timely preventive or corrective action [2].

Weather Parameters Analysed

Parameter	Purpose
Temperature	Influences pathogen growth and reproduction
Humidity	Promotes fungal development and disease spread
Rainfall	Increases leaf wetness and infection risk
Cloud Cover	Creates favourable conditions for pathogen survival
Wind Speed	Assists spore dispersal
Atmospheric Pressure	Supports weather pattern analysis

Table 10.1

Disease Risk Classification

The system generates a threat score between **0 and 100** and classifies the disease risk into five levels.

Score Range	Risk Level
0–29	Very Low
30–49	Low
50–69	Moderate
70–89	High
90–100	Critical

Table 10.2

Late Blight Monitoring and Recommendations

Favourable Conditions

- Relative Humidity above 85%
- Continuous rainfall
- Dense fog
- Cloudy weather
- Temperature between 10°C and 25°C [6]

Risk Level	Recommended Actions
Very Low	Continue routine monitoring, maintain field sanitation, and use certified disease-free seeds.
Low	Inspect crops twice weekly, improve field drainage, and remove infected debris.
Moderate	Begin preventive fungicide applications such as Mancozeb and Chlorothalonil.
High	Apply systemic fungicides such as **Ridomil Gold (700 g/acre)**, **Curzate M-8 (700 g/acre)**, or **Sectin (700 g/acre)**. Monitor fields daily.

Risk Level	Recommended Actions
Critical	Apply fungicides immediately, destroy heavily infected plants, conduct daily inspections, and coordinate treatment efforts to prevent disease spread.

Table 10.3

Recommended Resistant Varieties

- Kufri Jyoti
- Kufri Sutluj
- Kufri Badshah
- Kufri Chipsona-1
- Kufri Chipsona-2

Early Blight Monitoring and Recommendations

Favourable Conditions

- Temperature between 20°C and 30°C [6]
- Moderate to high humidity
- Nutrient deficiency
- Aging plant foliage

Risk Level	Recommended Actions
Very Low	Continue regular monitoring and maintain balanced crop nutrition.
Low	Perform weekly crop scouting, maintain field hygiene, and use certified seeds.
Moderate	Increase scouting frequency, improve air circulation, and reduce leaf wetness duration.
High	Begin fungicide treatment and monitor disease progression. Recommended products include **Luna Pro**, **Luna Tranquillity**, and **Velum Rise**.
Critical	Apply treatment immediately, remove infected foliage, and intensify field monitoring.

Table 10.4

Healthy Crop Monitoring

Even when a crop is classified as healthy, the system continues to analyse weather conditions and assess future disease risks.

Risk Level	Recommended Actions
Very Low	Continue normal farming practices.
Low	Monitor crop health regularly.
Moderate	Increase field inspections and check for early symptoms.
High	Begin preventive disease management practices and monitor crops daily.

Risk Level	Recommended Actions
Critical	Apply preventive fungicides and take immediate precautionary measures.

Table 10.5

Alert Generation System

The system automatically generates alerts and recommendations based on the calculated threat level.

Alert Information Includes:

- Disease type
- Threat score
- Risk level classification
- Critical risk period
- Alert title and message
- Action-required status

Key Benefits

- Early disease warning before visible symptoms appears.
- Weather-aware treatment and prevention recommendations.
- Reduced misuse of fungicides and pesticides.
- Improved crop yield and quality.
- Real-time decision support for farmers.
- Support for precision agriculture and sustainable farming [6].
- Enables proactive disease management through environmental monitoring.

By integrating weather forecasting, disease risk assessment, and treatment recommendations, the system provides farmers with a comprehensive decision-support tool for effective crop protection and disease management.

CHAPTER 11
CONCLUSION

Conclusion

This project successfully demonstrates the design and development of an intelligent potato leaf disease prediction system that combines image processing and deep learning to address one of the major challenges in modern agriculture— early and accurate disease detection [1]. By implementing a two-stage classification pipeline, the system first verifies whether an input image is a potato leaf and then accurately predicts the presence of Healthy, Early Blight, or Late Blight conditions.

The deep learning–based approach provides strong robustness against variations in lighting, background, and leaf orientation, ensuring reliable performance across diverse real-world scenarios [4]. The system's automation, high accuracy, and ease of use make it suitable for deployment in agricultural decision-support applications, enabling farmers to diagnose diseases quickly and take timely preventive measures.

Overall, this project demonstrates how integrating machine learning with agricultural practices can significantly improve crop monitoring, reduce manual workload, and enhance productivity. The solution developed serves as a scalable foundation for future enhancements, such as integrating more disease categories, real-time detection through mobile applications, and expanding the model to other crop species. Through this work, the project contributes meaningfully to the advancement of smart farming and sustainable agricultural technologies [2].

CHAPTER 12
FUTURE WORK

Future Work

1. Improving Model Accuracy

While the system achieves high accuracy in detecting potato leaves and predicting diseases, there is still room for improvement. Future work can focus on collecting more diverse leaf images from different regions, weather conditions, and growth stages. Training the model on larger datasets and including additional potato diseases can help improve its ability to handle real-world situations more effectively.

2. Integrating AI Assistance into the Application

In the future, an AI assistant can be integrated into the application to provide users with additional support. Besides predicting diseases, the assistant could explain symptoms, suggest possible treatments, answer farming-related questions, and provide guidance on disease prevention. This would make the application more informative and useful for farmers.

3. Enhancing User Experience

The user experience can be further improved by making the application more intuitive and accessible. Features such as multilingual support, offline prediction, disease history tracking, and personalized recommendations can make the system easier to use. A simpler interface and faster response times would also help farmers interact with the application more efficiently.

CHAPTER 13
BIBLIOGRAPHY

BIBLIOGRAPHY

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